Chessboard Problems — Solutions

Problem 1. (Bundeswettbewerb Mathematik 1978 Round 1 Problem 1) A knight is modified so that it moves p fields horizontally or vertically and q fields in the perpendicular direction (where $p, q \neq 0$). It is placed on an infinite chessboard. If the knight returns to the initial field after n moves, show that n must be even.

Solution. We identify the movement $(-p, -q)$ as -1 moves of $(+p, +q)$, etc. Let a, b, c, d be the numbers of moves of the form $(+p, +q), (+p, -q), (+q, +p), (+q, -p)$ respectively. Then we have

$$\begin{aligned} ap + bp + cq + dq &= 0, \\ aq - bq + cp - dp &= 0. \end{aligned}$$

WLOG assume $(p, q) = 1$.

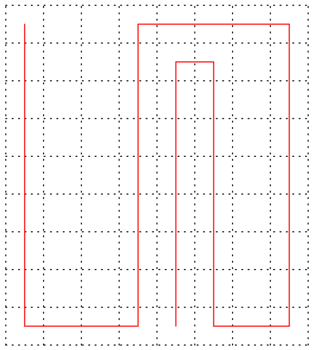
- If $p \equiv q \equiv 1 \pmod 2$, then the first equation implies $a + b + c + d \equiv 0 \pmod 2$. Thus, $n \equiv a + b + c + d \equiv 0 \pmod 2$.
- If $p \equiv 1 \pmod 2$ and $q \equiv 0 \pmod 2$, then the two equations imply $a + b \equiv 0 \pmod 2$ and $c - d \equiv 0 \pmod 2$. Thus, $n \equiv a + b + c + d \equiv 0 \pmod 2$.
- If $p \equiv 0 \pmod 2$ and $q \equiv 1 \pmod 2$, then the two equations imply $c + d \equiv 0 \pmod 2$ and $a - b \equiv 0 \pmod 2$. Thus, $n \equiv a + b + c + d \equiv 0 \pmod 2$.

Therefore, n is even in any case. $\square$

Problem 2. X and Y play a game on an $m \times n$ chessboard, where m and n are positive integers. Initially, the top left-hand corner cell is marked, and there is a rook in this marked cell. Starting from X, the two players take turn moving the rook to another cell in the same row or the same column such that all cells in between are unmarked (including the new cell that it stays in). Then all these cells in between are marked. The player who cannot move will lose the game. Find all m and n such that X has a winning strategy.

Solution. X has a winning strategy except $m = n = 1$.

The case $m = n = 1$ is trivial. WLOG assume $m \geqslant n$ and $m \geqslant 2$. The strategy of X is to move downwards and upwards alternately until the rook lies on the boundary. After the first step, Y has to move the rook to a boundary cell of the shorter side of an $m \times (n-1)$ rectangle. Afterwards, every time Y has to move the rook to a boundary cell of the shorter side of a rectangle because the length of the longer side is decreased by at most 1 and the length of the short side is decreased by at least 1 in each round (after X's move and Y's move). Then it is clear that X can always make a valid move. $\square$



Problem 3. (Tournament of Towns 2005 Fall Junior A Level Problem 3) Originally, every square of an 8×8 chessboard contains a rook. One by one, rooks which attack an odd number of others are removed. Find the maximum number of rooks that can be removed. (A rook attacks another rook if they are on the same row or column and there are no other rooks between them.)

Solution. At most 59 rooks can be removed.

There are 64 rooks in total. The 4 rooks at the corners can never be removed since they always attack exactly 2 other rooks. If all other rooks can be removed, we consider the moment when there are 5 rooks left. This includes the 4 rooks at the corners and another rook. If this rook lies on the boundary, then it attacks exactly 2 other rooks. If it lies in the interior, then it does not attack any other rook. In any case, we cannot remove this rook, contradiction. Therefore, at most 59 rooks can be removed. An example is given below, where the number indicates the order of removal.

X	1	8	15	22	29	36	X
38	2	9	16	23	30	50	44
39	3	10	17	24	31	51	45
40	4	11	18	25	32	52	46
41	5	12	19	26	33	53	47
42	6	13	20	27	34	54	48
43	55	56	57	58	59	X	49
X	7	14	21	28	35	37	X

□

Problem 4. (Baltic Way 2017 Problem 8) A chess knight has injured his leg and is limping. He alternates between a normal move and a short move where he moves to any diagonally neighbouring cell. The limping knight moves on a 5×6 chessboard starting with a normal move. What is the largest number of moves he can make if he is starting from a cell of his own choice and is not allowed to visit any cell (including the initial cell) more than once?

Solution. He can make at most 25 moves.

Except for the initial cell, the knight must visit a cell in the second row or the fourth row in one of every two consecutive steps, since he must move to a consecutive row in a short move. As there are 12 cells in these two rows, the number of cells to which the knight can visit is at most $1+2\times12+1=26$. In other words, at most 25 moves is possible. An example is given below, where the number indicates the order of the cells he visits.

1	10	21	23		26
3	8	11	20	22	24
5	2	9	14	25	
7	4	15	12	19	17
	6	13	18	16	

□

Problem 5. (Switzerland TST 2020 Problem 1) Let $n \geqslant 2$ be an integer. Consider an $n \times n$ chessboard with the usual chessboard colouring. A move consists of choosing a 1×1 square and switching the colour of all squares in its row and column (including the chosen square itself). For which n is it possible to get a monochrome chessboard after a finite sequence of moves?

Solution. n can be any even number.

For even n, we can choose all cells which have the same colour (say, black) as the cell in the top left-hand corner one by one. Then the colour of every black cell is changed $2\cdot\dfrac{n}{2}-1=n-1$ times, and so it becomes white. The colour of every white cell is changed $2\cdot\dfrac{n}{2}=n$ times, and so it remains white. Therefore, all cells are white at the end.

For odd n, let k be the number of black cells in the first two rows (which may change in each move). Initially, $k=n$ is odd. In each move, if we choose a cell in the first two rows, then the colour of $n+1$ cells in the first two rows are changed. Since $n+1$ is even, the parity of k is unchanged. If we choose a cell not in the first two rows, then the colour of 2 cells in the first two rows are changed. Again, the parity of k is unchanged. If the chessboard becomes monochrome, we must have $k=0$ or $k=2n$. Both are impossible since k is always odd. □

Problem 6. (Poland MO 2nd Round 2023 Problem 6) Given an $n\times n$ chessboard, where $n\geqslant 4$ and $p=n+1$ is a prime number. A set of n unit squares is called *tactical* if after putting down queens on these squares, no two queens are attacking each other. Prove that there exists a partition of the chessboard into $n-2$ tactical sets, not containing squares on the two diagonals. (Queens are allowed to move horizontally, vertically and diagonally.)

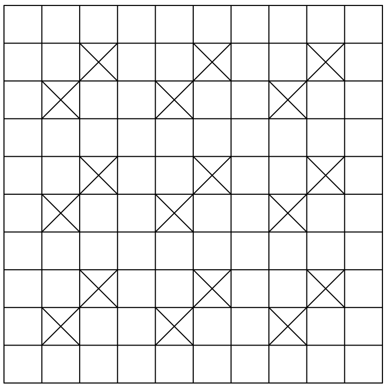
Solution. Let (i,j) be the cell in the ith row and the jth column. For each $k=2,3,\dots,n-1$, let S_k be the set of the cells (j,jk) where $j=1,2,\dots,n$ and the indices are taken modulo p (not n). We claim that $S_2,S_3,\dots,S_{n-1}$ is an admissible partition. The checking is given below.

- ((j, jk) is a cell in the $n \times n$ chessboard) This is because j and jk are not multiples of p.
- ((j, jk) is a cell not on the two diagonals) This is because $j \not\equiv \pm jk \pmod p$.
- ((j, jk) and $(j', j'k')$ are distinct cells) This is because $j \equiv j' \pmod p$ and $jk \equiv j'k' \pmod p$ iff $(j, k) = (j', k')$.
- ((j, jk) and $(j', j'k)$ do not lie in the same row if $j \neq j'$) Obvious.
- ((j, jk) and $(j', j'k)$ do not lie in the same column if $j \neq j'$) This is because $jk \equiv j'k \pmod p$ iff $j = j'$.
- ((j, jk) and $(j', j'k)$ do not lie in the same diagonal if $j \neq j'$) This is because $jk \pm j \equiv j'k \pm j' \pmod p$ iff $j = j'$.

$\square$

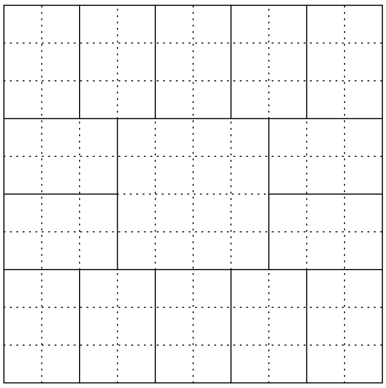
Problem 7. Some cells of a 10×10 chessboard are coloured blue. Determine the greatest value of n for which it is possible to colour n chessboard cells blue such that there is no 3×2 or 2×3 rectangle which is entirely blue.

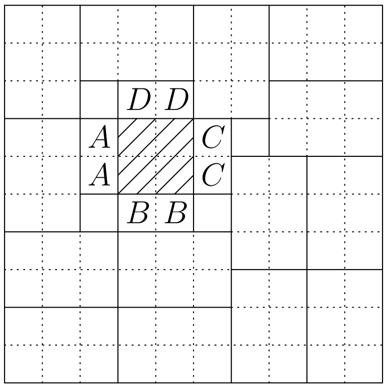
Solution. The greatest possible n is 82. The following gives a colouring with 82 blue cells, where a cross means a non-blue cell.



We now prove that there are at least 18 non-blue cells. Firstly, if there are at least 4 non-blue cells in the middle 4×4 square, we partition the chessboard in the way as shown in the left figure below. By the given condition, there is at least one non-blue cell in each of the 14 small rectangles in this partition. Thus, there are at least $4 + 14 = 18$ non-blue cells.

Secondly, if there are at most 3 non-blue cells in the middle 4×4 square, WLOG we may assume the top left 2×2 square of this middle square is entirely blue (the shaded cells in the right figure below). Then each 1×2 or 2×1 rectangle next to this 2×2 square must contain a non-blue cell. Also, by considering the partition as shown in the right figure below, there is at least one non-blue cell in each of the 14 small rectangles. Again, there are at least $4 + 14 = 18$ non-blue cells. This completes the proof.

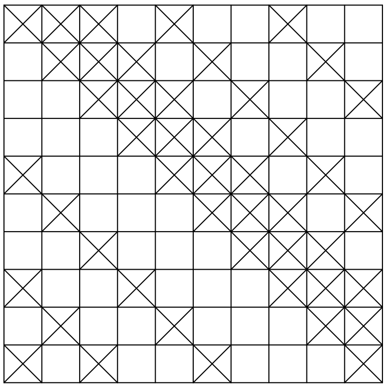




$\square$

Problem 8. (Brazil MO 2022 Problem 6) Some cells of a 10×10 chessboard are coloured blue. A set of six cells is called *gremista* when the cells are the intersection of three rows and two columns, or two rows and three columns, and are painted blue. Determine the greatest value of n for which it is possible to colour n chessboard cells blue such that there is not a gremista set.

Solution. The greatest possible n is 46. The following gives a colouring with 46 blue cells, where a cross means a blue cell.



We now prove that there are at most 46 blue cells. Consider the 7 columns with the most number of blue cells, say the first 7 columns. We remove the last 3 columns to obtain a 10×7 chessboard. Let $a_1, a_2, \ldots, a_{10}$ be the number of blue cells in the 10 rows respectively. Let x be the number of triples (i, j, k) with $1 \leqslant i < j \leqslant 7$ and $1 \leqslant k \leqslant 10$ such that the ith cell and the jth cell in the kth row are blue. On the one hand, for each k, there are $\binom{a_k}{2}$ ways to choose i and j. On the other hand, for each pair (i, j), there is at most 2 choices of k, or otherwise there is a gremista set. This yields

$$\binom{a_1}{2} + \binom{a_2}{2} + \cdots + \binom{a_{10}}{2} = x \leqslant 2\binom{7}{2} = 42.$$

Let $S = a_1 + a_2 + \cdots + a_{10}$. By Jensen's inequality, we have

$$42 \geqslant 10\binom{S/10}{2} = \frac{S(S-10)}{20}.$$

This implies $S \leqslant 34$. In other words, there are at most 34 blue cells in the first 7 columns. By the pigeonhole principle, one of these 7 columns contains at most 4 blue cells. Due to the choice of these 7 columns, the other 3 columns must contain at most 4 blue cells each. In total, there are at most $34 + 4 \times 3 = 46$ blue cells as desired. $\square$